

- 25** In cancer cells, the production of DNA ligase is upregulated, meaning it is produced in larger quantities than in a normal cell.

Several potential anticancer drugs are being designed as inhibitors to this enzyme as a mechanism for preventing DNA replication in cancerous cells.

How does the inhibition of DNA ligase prevent successful DNA replication in cancerous cells?

- A** Free nucleotides would be unable to form any phosphodiester bonds with adjacent nucleotides while forming the lagging strand.
- B** Hydrogen bonds between the nucleotide bases within the leading strand would **not** form, preventing complementary base pairing between the DNA strands.
- C** Short strands of DNA would **not** be successfully incorporated into the continuous sugar phosphate backbone of the leading strand.
- D** Condensation reactions between some nucleotides on the lagging strand would **not** be catalysed, preventing the nucleotides linking together.